

Question			Marking details	Marks available					Prac
				AO1	AO2	AO3	Total	Maths	
3	(a)	(i)	$n = \frac{\sin 50.0}{\sin 29.0} [= 1.58] \text{ or } \frac{v_{\text{air}}}{v_{\text{plastic}}} = \frac{\sin 50.0}{\sin 29.0} [\text{or by implication}] (1)$ $v = 1.90 \times 10^8 [\text{m s}^{-1}] (1)$	1	1		2	2	
		(ii)	39.3° Accept 39.2° and 39° ecf	1			1	1	
	(b)	(i)	$n_{\text{clad}} [\sin 90^\circ] = 1.530 \sin 81^\circ$ Give this first mark even if 9° instead of 81° (1) $n_{\text{clad}} = 1.51 (1)$		2		2	2	
		(ii)	$\frac{1}{\cos 9^\circ} \text{ or } \frac{1}{\sin 81^\circ}$		1		1	1	
		(iii)	Either starting from $\Delta t = 7.5 \text{ ns}$ $t_{\text{axial}} = \frac{\Delta t}{0.0125} [= 600 \text{ ns}] \text{ or by implication } (1)$ $d = \frac{ct_{\text{axial}}}{n} \text{ used } (1)$ $d = 118 \text{ m}$, so 150 m is too long. [Accept $d = 180 \text{ m}$ so 150 m not too long, arising from omission of n – already penalised] (1) or if starting from $d = 150 \text{ m}$, $t_{\text{axial}} = \frac{nd}{c} [= 765 \text{ ns}] \text{ or } t_{\text{zigzag}} = 775 \text{ ns} \text{ or by implication } (1)$ $\Delta t = 0.0125 t_{\text{axial}}$ or equivalent used (1) $\Delta t = 9.6 \text{ ns}$, (accept 10 ns) so 150 m is too long. [Accept $\Delta t = 6.25 \text{ ns}$ so 150 m not too long, arising from omission of n , already penalised.] (1)			3	3	3	
			Question 3 total	2	4	3	9	9	0